Programming Methods Used in Commercial OBD Devices

Overview

Modern OBD (On-Board Diagnostics) devices used in vehicles require reliable programming methods both during manufacturing and after deployment.

Most companies use a combination of hardware-based programming during production and wireless updates in the field.

1. Pogo-Pin Programming (Manufacturing Stage)

Pogo-pin programming is the industry-standard method used in factories. Instead of USB or exposed connectors, manufacturers place test pads

on the PCB. A pogo-pin jig (bed-of-nails) makes temporary contact to flash firmware.

Why Companies Use Pogo Pins

• Fast flashing (seconds per unit)

• No connector cost or wear

• Easy automation for mass production

• Secure (no user access after sealing)

Typical signals exposed via test pads:

EN, IO0, TX, RX, GND, 3.3V

2. OTA (Over-The-Air) Programming (Post-Deployment)

Once the OBD device is installed in a vehicle, physical access is impractical. OTA updates allow firmware upgrades remotely using cellular connectivity.

Why OTA is Mandatory

• Bug fixes without recalling vehicles

• Feature updates

• Security patches

• Configuration changes

Professional OTA Implementation

Commercial devices use a dual-partition (A/B) firmware strategy with rollback protection. Firmware images are signed, verified, and only activated

after integrity checks.

What Companies Avoid

• Exposed USB ports

• User-accessible UART pins

• Manual reflashing after deployment

• Unsafe OTA without rollback support

Recommended Architecture

• Pogo-pin programming for factory flashing and testing

• OTA updates for field upgrades

• Locked bootloader after production

• Secure backend-controlled firmware delivery

Conclusion

Real-world OBD and telematics products always support both methods. Pogo-pin programming ensures efficient manufacturing, while OTA ensures

long-term maintainability and scalability of deployed devices.